

Resampling

practical limitations...

- often we don't know the distribution of the random variable
- often samples are not drawn at random

with time and computational power, **empirical distributions** can be built by **resampling** data, and used for

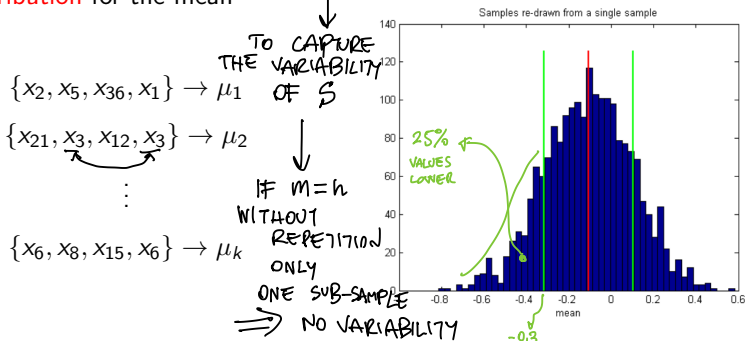
- confidence intervals
- hypothesis testing
- ...



INTERMEDIATE BETWEEN
DISTRIBUTION OF VALUES
AND PROBABILITY DISTRIBUTIONS

bootstrap

- we want to estimate the mean of a random variable with **unknown distribution**
- a n -dimensional sample $S := \{x_1, x_2, \dots, x_n\}$ is observed
 $\rightarrow K = 1000$ OR 10000
- many m -dimensional ($m \leq n$) sub-samples **with repetition** are extracted from S and the mean is observed on each subsample to create an **empirical distribution** for the mean
 $\rightarrow$ REPLACEMENT



- empirical distribution used to estimate confidence intervals or testing hypotheses

-0.3 IS THE QUANTILE ASSOCIATED TO THE 25% PERCENTILE

exercise 1.6

reshuffling

TELLS HOW MUCH TWO VARIABLES VARY TOGETHER

- investigating the correlation of two random variables X and Y
- Pearson's correlation coefficient $\rho_{XY} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$ → LINEAR CORRELATION:
 $Y = AX$ $\rho_{XY} \in [-1, 1]$
- breaking the association structure between X and Y by randomly reshuffling many times one of the two samples and obtaining empirical distribution of the correlation to see if it is reliable $\rho \approx 1$ → POSITIVE CORR.

	height	weight
person 1	1.875714232	109.8196777
person 2	1.747060363	73.68895452
person 3	1.882396677	96.58434842
person 4	1.821966851	99.89928152
person 5	1.774997615	93.68280948
person 6	1.708226598	69.10431871
person 7	1.747141064	83.50326143
person 8	1.736052294	76.25888416
person 9	1.702281321	79.87196594
person 10	1.611794947	71.00545308
person 11	1.80836271	84.7186362
person 12	1.81967645	97.03849095
person 13	1.64506476	75.87586733
person 14	1.75978998	86.00856635
person 15	1.758790799	84.64111228
person 16	1.71819874	78.17286625
person 17	1.839425242	88.99694187
person 18	1.624947873	78.48909547
person 19	1.768857521	84.43671674
person 20	1.725574523	82.8216982
corr		0.818411356



height	weight	height	weight	height	weight
1.875714232	75.88	1.875714232	79.87	1.875714232	109.82
1.747060363	96.58	1.747060363	83.5	1.747060363	86.01
1.882396677	84.64	1.882396677	96.58	1.882396677	89
1.821966851	79.87	1.821966851	71.01	1.821966851	99.9
1.774997615	97.04	1.774997615	78.17	1.774997615	76.26
1.708226598	76.26	1.708226598	73.69	1.708226598	96.58
1.747141064	69.1	1.747141064	82.82	1.747141064	97.04
1.736052294	93.68	1.736052294	93.68	1.736052294	84.44
1.702281321	73.69	1.702281321	75.88	1.702281321	78.49
1.611794947	84.72	1.611794947	84.44	1.611794947	84.64
1.80836271	82.82	1.80836271	84.72	1.80836271	71.01
1.81967645	86.01	1.81967645	78.49	1.81967645	75.88
1.64506476	89	1.64506476	86.01	1.64506476	78.17
1.75978998	83.5	1.75978998	76.26	1.75978998	73.69
1.758790799	99.9	1.758790799	69.1	1.758790799	93.68
1.71819874	78.17	1.71819874	109.82	1.71819874	82.82
1.839425242	78.49	1.839425242	89	1.839425242	84.72
1.624947873	84.44	1.624947873	84.64	1.624947873	69.1
1.768857521	71.01	1.768857521	99.9	1.768857521	83.5
1.725574523	109.82	1.725574523	97.04	1.725574523	79.87
	-0.129865955		-0.039975309		0.384984573

$\rho \approx -1$

→
NEGATIVE
CORR.

$\rho \approx 0$

→
NO LINEAR
CORR.

IF CORRELATION ρ IS NOT DUE TO LOW DATA VARIABILITY

exercise 1.7



Overview

- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis**
- 4 Unsupervised learning: Clustering
- 5 Supervised learning
- 6 Overview on Time Series

Multivariate statistics

some important questions...

- what data is?
- how we look at data?
- how we manipulate data?
- how we find things in data?

different combination of alternatives are possible!

SELF DRIVEN CARS

- IMAGES FROM EXTERNAL AMBIENT
- SENSORS COLLECTING IMAGES
- TRANSFORM PIXELED IMAGES INTO NUMERICAL MATRICES
- DEEP LEARNING

MEDICAL DIAGNOSIS

- CLINICAL DATA OF PATIENTS
- CLINICIANS COLLECTING CLINICAL DATA
- NORMALIZE DATA IN $[0,1]$
- SIMPLE MACHINE LEARNING MODEL (NEURAL NET.)



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data is composed by

↑ DENOTES THE SINGLE
OBSERVATION

Rows • individuals/observations/units/samples

Columns • features/characteristics/variables/attributes

minimal data requirements

- distinguishable individuals
- measurable features

many individuals, many features $\Rightarrow$ multivariate statistics

some assumptions...

WE MAY ALSO
HAVE CATEGORICAL
FEATURES

- features can be measured on a continuous scale
- data is represented in a grid (matrix):
rows = individuals
columns = features
- matrix element (i, j) is the measure of feature j on individual i
- m is the number of rows (individuals)
- n is the number of columns (features)

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

$$x_i = \begin{pmatrix} x_{1i} \\ x_{2i} \\ \vdots \\ x_{mi} \end{pmatrix}$$

$$X = (x_1, x_2, \dots, x_n)$$

...data has been put into a "space" ...

what do you see?

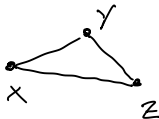
OBSERVE THE DISTANCE BETWEEN
OBJECTS !!!



data represented in a **space** $\Rightarrow$ distance (metric) d

$\nearrow$ INPUT: A PAIR OF INDIVIDUALS FROM X
 $d : X \times X \rightarrow [0, \infty)$ properties $\rightarrow$ OUTPUT: DISTANCE

- ① $d(x, y) \geq 0$ (non-negativity)
- ② $d(x, y) = 0 \iff x = y$ (identity of indiscernable)
- ③ $d(x, y) = d(y, x)$ (simmetry)
- ④ $d(x, y) + d(y, z) \geq d(x, z)$ (subadditivity)



exercise 2.1